

account for the fact that many planets appear to turn back in their orbits, performing loops in the sky that Ptolemy called “epicycles,” or orbits within orbits. Copernicus set out to explain this, and his answer showed that the planets do not turn back in their orbits, but only appear to do so as the Earth overtakes them in their journey around the Sun.

Breaking with convention

Copernicus had gained a reputation as a highly adept astronomer, and in 1514, he privately circulated copies of a handwritten pamphlet that proposed an alternative theory of planetary motion. He placed the Sun at the center of the Universe, with the Earth and all the planets orbiting it in a fixed order. Further, he argued that the Earth spins on its axis, accounting for the observed daily movements of the planets, the stars, the Moon, and the Sun. Finally, he proposed that the stars lie far beyond the farthest planet, suggesting that the Universe must be much larger than had previously been believed.

This revelatory view of the solar system solved many—but not all—of the problems with the geocentric model. Copernicus himself had trouble with it and was forced to follow Ptolemy’s example and include epicycles in order to account for certain phenomena. (He wrongly believed that the planets move in circular rather than elliptical orbits.)

However, despite its imperfections, Copernicus’s Sun-centered, or heliocentric, model of the Universe (*helios* meaning “Sun” in Greek) greatly outstripped the Ptolemaic model in terms of accuracy and gradually gained credence in intellectual circles. Georg Joachim Rheticus, a German

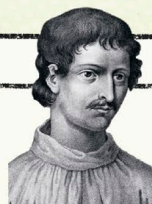
mathematician and pupil of Copernicus, strongly encouraged him to publish his theory, but he only did so in 1543, as he was dying. His seminal work was published as *De Revolutionibus Orbium Coelestium* (*On the Revolutions of the Celestial Spheres*). It contradicted the Church’s teachings but was only branded heretical in 1616, when it was endorsed by Italian astronomer Galileo Galilei (see pp.50–55).

Posthumous influence

Providing a radical new view of the Universe and challenging centuries of received wisdom, Copernicus’s controversial work paved the way for the Scientific Revolution and the rebirth of philosophy in Europe. It marked the end of an old idea—that humans stand at the center of things—and set up a dispute between science and religion that continues to this day.

GIORDANO BRUNO

An Italian friar and astronomer, Bruno was executed for supporting the heliocentric model.



Part of the Dominican order, Bruno (1548–1600) was a radical thinker who rejected the Church-endorsed geocentric view of the Universe. Instead, he supported Copernicus’s heliocentric model and took the theory further by suggesting the Universe was infinite and might have numerous worlds within it. His unorthodox views angered Church authorities, who tried him for heresy. He was burned at the stake in 1593.

